INCH-POUND

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MILITARY SPECIFICATION

LIFT, VEHICLE DRIVELINE COMPONENTS

(2000 POUND CAPACITY)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a manually operated hydraulic lift of 2000 pound capacity for removing and replacing transmissions, transfer cases, and differentials of military and commercial wheeled vehicles.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (6.2.1).

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-T-60 - Tape, Packaging, Waterproof.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4910
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MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-H-6083	- Hydraulic Fluid, Petroleum Base, for Preservation and Operation.
MIL-F-17111	- Fluid, Power Transmission.
MIL-P-19834	- Plates, Identification or Instruction, Metal Foil, Adhesive Backed, General Specification for.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods.
MIL-STD-1189	- Bar Code Symbology, Standard Department of Defense.
MIL-STD-1190	- Minimum Guidelines for Level C Preservation, Packaging, Marking.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2.1).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E4	- Standard Practices for Load Verification of Testing Machines.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subject to first article inspection (see 6.4) in accordance with 4.2.1. The approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The vehicle driveline component lift shall be a manually operated hydraulic lift of 2000 pound capacity for removing and replacing transmissions, transfer cases, and differentials of military and commercial wheeled vehicles. The lift shall have an internal load limiting device conforming to 3.6.6.

3.3 Parts and materials. Parts and materials used in the fabrication of the lift shall be new and free of defects that adversely affect the safety, performance, or serviceability of the finished product. Contact between dissimilar metals, especially aluminum or magnesium in contact with steel, copper, or brass shall be avoided whenever practical. When contact between dissimilar metals is required by design, the metals shall be protected against galvanic corrosion in accordance with MIL-STD-889.

3.4 Construction. Lifts shall be complete so that they can be used for the operations specified herein.

3.4.1 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for adjustment and for preventing loss of tightness. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

3.4.2 Castings and forgings. Castings and forgings shall be free of surface defects, scale, and mismatching. Defective castings or forgings shall not be reclaimed for use in fabrication of the lift.

3.4.3 Welding, brazing, and soldering. Welding, brazing, and soldering shall be employed only where specified in the original design. These operations shall not be employed as a repair measure to reclaim defective parts for use in the lift. Welding, brazing, and soldering shall conform to the best commercial practice and shall provide the mechanical strength and durability required by the performance limits established herein.

3.4.4 Surfaces. Surfaces of all castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free of sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials.

3.4.5 Threads Threaded parts used on the lift shall conform to FED-STD-H28 and the applicable detailed Standard section referenced therein.

3.4.6 Lubrication. Bearings, except sealed-for-life type, and all other working parts requiring periodic lubrication shall be provided with accessible means for lubrication. Each lift shall be properly lubricated when delivered to the Government.

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3.4.7 Tie down equipment. Two chains, webbed slings, or wire ropes, each eight feet long and rated for 500 pounds or more working load, shall be provided for securing the load during transporting, lifting, and lowering. Each chain, web, or wire rope shall be furnished with a positive tensioning device rated for 500 pounds or more that can be attached at any position on the chain, web, or rope. Wire ropes, when used, shall have a flemish eye formed in each end. The flemish eye shall be large enough to slip either eye through the other eye to form a sling. Webbing shall be ultra-violet resistant, abrasion resistant, snag resistant, oil resistant, and water resistant. Webbing ends and edges shall be woven, or otherwise protected, to resist fraying and unraveling.

3.4.8 Caution plates. A caution plate shall be marked in a legible and permanent manner in accordance with 3.4.8.1 and 3.4.8.2, shall be made of either of the following materials:

(a) Adhesive backed aluminum foil plate conforming to MIL-P-19834.
(b) Adhesive backed plastic or plastic-coated plate, having physical characteristics equivalent to the aluminum foil plate specified above, that complies with the tests specified in MIL-P-19834.

3.4.8.1 Fluid identification plate. The manufacturer shall indicate either MIL-H-6083 or MIL-F-17111 as the applicable hydraulic fluid to be used for refilling the jack (see 3.6.3.2). If the lift used hydraulic fluid meeting the requirements of MIL-F-17111, the following message shall be permanently and legibly applied to a plate conforming to 3.4.8. The plate shall be permanently attached in a conspicuous location near the filler plug(s) of the reservoir.

CAUTION
USE NO FLUID OTHER THAN
MIL-F-17111

If the lift uses hydraulic fluid meeting the requirements of MIL-H-6083, the following message shall be permanently and legibly applied to a plate conforming to 3.4.8. The plate shall be permanently attached in a conspicuous location near the filler plug(s) of the reservoir.

CAUTION
USE NO FLUID OTHER THAN
MIL-H-6083

3.4.8.2 Safety warning plate. The following message shall be permanently and legibly applied to a caution plate in accordance with 3.4.8. The plate shall be permanently attached to the lift in a conspicuous location visible to the operator.

WARNING

THE USE OF THIS LIFT IS LIMITED TO
THE INSTALLATION, REMOVAL, AND TRANSPORTATION
OF TRANSMISSIONS, TRANSFER CASES AND DIFFERENTIALS

DO NOT
LIFT OR SUPPORT VEHICLE
WITH THIS LIFT

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3.4.9 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such part of a particular type or model can be replaced or interchanged without modification of the replaced part or any other part of the lift.

3.5 Performance and product characteristics.

3.5.1 Functional requirements. The lift shall safely lower, convey, lift, position for installation, and sustain vehicle transmissions, transfer cases, and differentials within the weight capacities specified in 3.5.2 and as identified in 3.5.5.1.

3.5.2 Rated load capacity and retention. The lift shall have a rated load capacity of no less than 2000 pounds. The lift and its hydraulic system shall retain the full rated load capacity in all positions between the upper and lower limits of the lifting range. The lift shall hold 2000 pounds with the lifting arm in the horizontal position for four hours with no more than .0625 inch drop in the height of the lift plate. The lift shall also hold 2000 pounds at 50 percent, 75 percent, and 100 percent of the lifting range with no more than .008 inches drop in a half hour period for each position.

3.5.3 Proofload. The lift, with the internal load limiting device deactivated, shall be capable of lifting and sustaining at all positions within the lifting range of 3.5.4 for 5 minutes without a drop of more than .03 inch, a proofload of 125 percent of rated capacity without permanent deformation or fracture of any component, hydraulic leakage, or binding of the lifting mechanism.

3.5.4 Support plate positioning. The height position of the support plate shall be adjustable within a lifting range between 10 inches or lower from the floor and 33 inches or higher from the floor when measured at the center of the support plate upper surface while the support plate is in a horizontal position $\pm$ 2 degrees. The support plate shall be tiltable from the horizontal plane from 0 degrees to no less than 15 degrees both forward and backward and to both sides when loaded with the rated load of 3.5.2 in all height positions except for the lowest positions in which the lifts base may interfere with tilting. The lift shall remain stable when the rated load is elevated to its greatest height and tilted to its maximum degree of tilt in all four directions. The support plate shall be capable of maintaining the vehicle component's attitude $\pm$ 2 degrees throughout the entire height range.

3.5.5 Support plate and truck component handling system. A driveline component handling system shall be provided with each lift that shall interface with and retain all military truck test components and any vehicle drive line components as to size, i.e., height, width, length, on an infinitely adjustable basis within the various configurations specified herein (see 3.5.1). The handling system shall have no noticeable permanent deformation at the rated load capacity (see 3.5.2) and at the proofload (see 3.5.3). Instructions for the proper use of all brackets and adapters of the vehicle component handling system shall be delivered with each lift. The instructions shall depict the proper application of the brackets and adapters to all of the vehicle components listed in 3.5.5.1, and as specified in 3.5.5.

3.5.5.1 Military vehicle list and driveline components. The lift shall be capable of lifting the following military vehicle driveline components.

A. M816 WRECKER 5 TON 6x6

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Transmission manufacturer (mfg)	Dana
Transmission model	6453
Transmission weight	421 pounds
Transmission height from floor plan	30 inches
Transfer case mfg	Rockwell
Transfer case model	T 138
Transfer case weight	660 pounds
Transfer case height from floor	24 inches

B. M936 5 TON TRUCK

Transmission mfg	Detroit Diesel Allison
Transmission model	MT 654CR
Transmission weight	90 pounds
Transmission height from floor to pan	26 inches
Transfer case mfg	Rockwell
Transfer case model	T 1138
Transfer case weight	578 pounds
Transfer case height from floor	23 inches

C. M35A2, 2 1/2 TON TRUCK, 6x6

Transmission mfg	Spicer
Model	WN-7-28
Transmission weight	250 pounds
Transfer case mfg	Rockwell
Transfer case model	T 136 27
Weight	450 pounds

D. M919 CEMENT TRUCK

Transmission mfg	Caterpillar
Transmission model	D7155/Semi Automatic
Transmission weight	1090 dry
Transmission height from floor to pan	30 inches
Transfer case mfg	Oshkosh
Transfer case model	1800 series F-U29
Transfer case weight	540 pounds
Transfer case height from floor	12 inches
Intermediate axle differential carrier	Detroit Automotive
Type	No Spin

E. M915A1 DIESEL TRACTOR

Transmission mfg by	Detroit Diesel Allison
Transmission model/PN	23014691
Transmission weight	1350 pounds
Transmission height above floor	Unknown

F. HEMTT

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Transmission case mfg	Oshkosh
Transmission case model	55000
Transmission case weight	882 pounds
Transfer case height from floor	27 inches
Height of transfer case	39 inches

3.5.6 Controls. Raising, lowering, and tilt controls shall be accessible to permit the operator to operate the lift and tilt controls and guide the vehicle component to the mating surface on the vehicle during removal or replacement of the vehicle component while laying on the floor beside the lift under the vehicle. There shall be only one control knob for each direction of movement for the support plate; the pump handle shall raise the support plate, the release lever shall lower the support plate, one control shall tilt the support plate side to side, and one control shall tilt the support plate forward and backward. It shall not be permissible to use pairs of adjusting controls that require two separate operations to gain the controlled motion in one direction of tilt. Hexagonal protrusions suitable for fitting standard socket wrenches, open end or box wrenches shall be provided along with hand grip knobs for adjusting support plate tilt. The hand knobs shall be for use when there is no load on the support plate and the hex protrusions shall be for use when there is a load on the support plate. The knob and hex protrusion shall both drive the same adjusting mechanism, respectively. A handle shall be located at each of the four lower corners of the lift, near the wheels and casters, that will aid in the controlled movement of the fully loaded lift.

3.5.7 Mobility and stability. The lift, loaded with no less than the rated load at the manufacturer's recommended height or higher for transporting loads, shall be capable of being moved in all directions over hard floor surfaces and shall be capable of traversing up onto, across, and down off of a fixed rigid obstacle approximately five feet wide consisting of a 0.5 $\pm$ 0.03 inch high 15 $\pm$2 degree slope and subsequent drop to floor level while being moved at a rate of 15 or more feet per minute. The lift shall traverse the rise at an approach angle which will bring only one caster at a time in succession onto and off of the rise. The subsequent mobility of the lift shall not be impaired as a result of the traverse and there shall be no evidence of damage to any components, fluid leakage or seepage, or binding of the lift mechanism. The lift and the load shall remain stable during the traverse. The lift, loaded with no less than the rated load at the manufacturer's recommended height or higher for transporting loads, shall be capable of striking simultaneously with any two adjacent casters, a 4 $\pm$ 0.125 inch vertical fixed rise in the floor while traveling at 15 or more feet per minute. The lift shall not lose the load, tip, become unstable, or permanently deform as a result of striking the 4 inch vertical rise.

3.6 Components.

3.6.1 Frame. The frame shall be capable of sustaining the loads specified in 3.5.2 and 3.5.3 without permanent deformation or binding.

3.6.2 Casters. The lift frame shall be fitted with no less than four swivel casters. The casters shall be ball or roller bearing type. The bearings shall be held in races, retainers, or cast recesses. When loaded to the proofload capacity, all casters shall rotate and swivel freely and the axles and casters shall not deform. Casters shall be replaceable and shall not loosen or tighten under service conditions.

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3.6.3 Hydraulic lifting mechanism. The lift shall incorporate a manually operated hydraulically powered lifting mechanisms for raising and lowering the lifting member. The lifting mechanism shall raise the rated load to any point throughout the specified lifting range with no more than 70 pounds force applied to the pump lever in the direction of gravity with the lever at 30 degrees from the horizontal. The lift shall incorporate a method to prevent further actuation of the lifting mechanism when the lifting member has reached its greatest travel. The unrestricted build-up of hydraulic pressure after the travel limit has been reached is not allowed.

3.6.3.1 Hydraulic unit. The hydraulic unit shall be self-contained and capable of raising, sustaining, and lowering the proofload (see 3.5.3) throughout the specified lifting range without leakage or binding.

3.6.3.2 Hydraulic fluid and seals. Only hydraulic fluid conforming to MIL-F-17111, or MIL-H-6083 shall be provided in each lift. The seals of the lift's hydraulic unit shall be compatible with the hydraulic fluid chosen from the above by the lift manufacturer and used in the jack. The solid particle contamination requirements in MIL-H-6083 shall not apply for hydraulic fluid used in these lifts. Instead, the hydraulic fluid shall be filtered through a 15 micron or finer mesh screen prior to packaging by the fluid manufacturer.

3.5.3.3 Pump lever. The pump lever shall rotate its pumping position through 200 degrees of more.

3.6.4 Reservoir. The reservoir shall have no less than 10 percent surplus capacity in excess of the quantity of hydraulic fluid required by the ram to reach its greatest extension. The reservoir shall have a filler plug(s) and shall be filled at the time of delivery to the Government with hydraulic fluid conforming to MIL-F-17111 or MIL-H-6083 (see 3.6.3.2).

3.6.5 Control valve. The manually activated hydraulic release system shall provide controlled lowering and complete stopping of the fully loaded and completely unloaded lift arm and platform at any position throughout its descent from its highest possible position to its lowest possible position. The hydraulic release system shall provide a controlled rate of descent to one foot or less per minute, from the highest position to the lowest position, without stopping, while under full capacity load. Under no load, the lift arm and platform shall lower from top to bottom in 30 seconds or less. All lowering rate requirements shall be at ambient temperature.

3.6.6 Load limiting device. A hydraulically activated internal load limiting device shall be provided in the pump side of the hydraulic circuit which actuates at loads greater than the rated capacity but less than 125 percent of the rated capacity of the lift. The hydraulically activated load limiting device shall require special tools or access techniques for adjustment and deactivation and shall be separate in construction from the load lowering control valve.

3.7 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting.

3.8 Identification marking. The lift shall be permanently marked for identification in accordance with MIL-STD-130 (applying the words, U.S. Government, is not required) and shall include the National Stock Number. The rated load

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capacity shall be permanently and legibly marked on a nameplate conforming to MIL-P-19834 and shall also be stamped or cast into the lift frame in characters no less than 0.374 inches high.

3.9 Workmanship. The lift shall be free of defects which adversely affect function or appearance. All parts except the adapters, tie downs, and pump handles (see 5.1.1.3 and 5.2.3) shall be assembled, mounted, and secured to the lift so as to accomplish the purpose for which intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements of the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operation is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective materials, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. When specified (see 6.2.1), first article inspection shall be applied to the preproduction model(s) or initial production item(s) noted in 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination specified in 4.3 and all tests specified in 4.4. Failure of the first article to pass the examination of any of the tests shall be cause for rejection.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to production items offered for acceptance under the contract. Quality conformance inspection shall consist of (a) through (e) as follows. Failure of any unit to pass an examination, test, or inspection shall be cause for rejection.

(a) Examination (see 4.3).
(b) Proofload test (see 4.4.2).
(c) Load limiting device test (see 4.4.3).
(d) Load retention and release test (see 4.4.4).
(e) Packaging inspection (see 4.5).

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4.3 Examination. Visually, manually, and dimensionally examine the lifts to determine conformance with the requirements of 3.3 through 3.4.8.2, 3.5.5, 3.5.6, 3.6 through 3.6.4, and 3.7 through 3.9. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the lift and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item (see 3.4.9) shall be cause for rejection.

4.4 Tests.

4.4.1 Test equipment.

4.4.1.1 Standard portable loads. Two standard portable loads for testing purposes shall be provided by the contractor. The portable loads shall each consist of a steel cylinder, three feet long and two feet in diameter, filled with lead or other solids as needed to achieve the desired weight. One of the standard portable loads shall weigh not less than the lift's rated load of 2000 pounds, and the other shall weight not less than the lift's proofload of 2500 pounds. Prior to use, the weight of the portable loads shall be verified on a scale accurate and readable to the nearest twenty pounds; and the loads shall be marked to indicate the location of the center of balance.

4.4.1.2 Compression testing machine. At the option of the contractor, the loads specified in the tests of 4.4.2 through 4.4.4 may be applied using a compression testing machine for quality conformance inspection. The testing machine shall incorporate alignment devices which ensure that the load is applied axially, and uniformly; and shall provide a constant force on the transmission lift support plate through its full lifting range, regardless of whether the support plate is being raised, lowered, or held stationary. The accuracy of the compression testing machine shall be verified in accordance with ASTM E4. The permissible variation for the testing machine is $\pm$1.0 percent, within a verified loading range of 2000 to 2500 pounds.

4.4.1.3 Truck Component Handling System. Applicable parts of the Truck Component Handling System supplied with the lift shall be used in addition to the tie down equipment (see 3.4.7) to secure the standard portable load to the lift.

4.4.2 Proofload test. With the load limiting device deactivated and the support plate level and at its lowest position, the lift shall be operated to raise the proofload (see 3.5.3) until the support plate is fully raised. While raising the load, the lift shall support the proofload to the extent that there is no visible dropping of the load. Failure of the lift to raise the support the proofload without damage or visible permanent deformation of the lift; binding of the lift mechanism; or leakage of the lift's hydraulic system shall be cause for rejection (see 3.5.3). For first article testing only, the lift shall be moved about while carrying the proofload. Failure of the casters to roll and swivel without binding while carrying the proofload shall be cause for rejection (see 3.5.3).

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4.4.3 Load limiting device test. The load limiting device shall be adjusted to activate at a load greater than the lift's rated load capacity, but less than 1.25 times the rated load capacity, in accordance with 3.5.2 and 3.6.6. The lift shall then be operated to raise a load equal to the rated load, and to attempt to raise a load equal to 1.25 times the rated load. Failure of the load limiting device to enable the lift to raise a load equal to the rated load, or to prevent the lift from raising a load equal to 1.25 times the rated load shall be cause for rejection.

4.4.4 Load retention and release test. With the load limiting device activated and adjusted in accordance with 3.5.2 and 3.6.6, the proofload shall be applied to the lift at the lift's maximum lifting height. The lift's release system shall be adjusted to fully and continuously lower the load at the lift's slowest rate of descent; and the rate measured. The proofload shall then be removed and reapplied to the lift at the lift's maximum lifting height, and the lift's release system adjusted to lower fully and continuously lower the load at the lift's fastest rate of descent and the rate measured. The proofload shall then be removed and reapplied to the lift at the lift's maximum lifting height, and the release system manipulated to lower and stop the load at points equal to 50 and 25 percent of the lifting height range of the lift, and the lift tested in the following manner at each point:

> The distance between the support plate and the floor shall be measured with a height gage having an accuracy of at least $\pm$0.001 inch. The lift shall then be allowed to stand for 5 minutes. The height of the support plate shall be measured again at the end of the 5 minute period.

The support plate shall then be fully raised and lowered while under no load. Failure of the release system to smoothly and safely control the rate of descent of the proofload to 0.25 inch or less per second at its slowest setting; failure of the release system to smoothly and safely control the rate of descent of the proofload to not more than 1.50 inch per second at its fastest setting; failure of the support plate to fully and smoothly lower under no load, failure of the release system to stop the load at any point in its descent and maintain the load within 0.03 inch of the stopping point for 5 minutes, or evidence of leakage shall be cause for rejection (see 3.5.3 and 3.6.5).

4.4.5 Force on handle test. The lift shall be operated to raise its rated load until the lifting member is in the horizontal position. The pump handle shall then be positioned at 30 degrees to the horizontal, and the force needed to move the handle and raise the rated load shall be measured by applying force straight downward at the end of the pump handle. Failure of the lift handle to be moved to the bottom of its stroke by a downward force of 70 pounds or less shall be cause for rejection (see 3.6.3).

4.4.6 Support plate test. The rated load shall be secured to the lift support plate and brackets with the plate level and raised not more than six inches above the wheels. With the rated load in place, the support plate shall be adjusted for maximum tilt to the front, rear, and each side, in any sequence. The lift shall then be operated to raise the load at maximum forward tilt until the support plate is in its highest position, and the sequence of tilts repeated. Inability of the loaded support plate to provide at least 15 degrees of tilt in each direction, failure of the support plate to maintain the attitude of the load throughout the lift, tipping of the lift, loss of the load, evidence of permanent deformation or breakage of the lift or truck component handling system, binding of the lift mechanism, or lift hydraulic system leakage shall be cause for rejection (see 3.5.4 and 3.5.5).

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4.4.7 Mobility test. The rated load shall be secured to the lift support plate, and the plate adjusted to the manufacturer's recommended height for transporting loads. While loaded, the ability of the lift to roll in all directions over a level floor shall be verified. The lift shall then be rolled onto, across, and off of an obstacle consisting of a 15 ±2 degree slope to a height of 0.50 ±0.03 inch immediately followed by a perpendicular drop to floor level, at a speed of 15 or more feet per minute. The lift shall traverse the rise of an approach angle which brings only one caster at a time onto and off of the obstacle. Failure of the lift to negotiate the obstacle as evidenced by its stopping, tipping over, losing the load, or sustaining damage (including permanent deformation, breakage, binding of the lift mechanism, and hydraulic system leakage), shall be cause for rejection (see 3.5.7).

4.4.8 Stability test. The rated load shall be secured to the lift support plate, and the plate adjusted to the manufacturer's recommend height for transporting loads. While loaded, the lift shall be pushed at a speed of 15 or more feet per minute so that two adjacent casters of the lift simultaneously strike a solid fixed obstacle consisting of a 4.000 ±0.125 inch vertical rise. The personnel performing the test shall stand clear of the lift and its load at the time of impact. No external constraints shall be applied to the lift or the load to prevent the lift from tipping. The test shall be repeated for each pair of adjacent casters in turn. Failure of the lift to withstand the impact as evidenced by its tipping over, losing the load, or sustaining damage (including permanent deformation, breakage, binding of the lift mechanism, and hydraulic system leakage) shall be cause for rejection (see 3.5.7).

4.4.9 Truck component handling test. The lift, in conjunction with its truck component handling system and tie down equipment, shall be used to remove, lower, raise, position, and replace the transmission and transfer case in Item B of 3.5.5.1, the intermediate axle differential carrier in Item D, and the transfer case in Item F. For the transport portion of this test, the lift shall be used to move each component removed from a vehicle, a total distance of at least 100 feet in the immediate vicinity of its vehicle. The ability of the lift and its truck component handling system and tie down equipment to attach to and firmly secure the remainder of the components listed in 3.5.5.1 shall be demonstrated. All vehicles, vehicle components, labor and tools will be supplied by the Government. Failure of the lift, its truck handling system, or its tie down equipment to mate securely to each of the components listed in 3.5.5.1 failure of the lift, its truck handling system, or its tie down equipment to facilitate removal, lowering, transport, raising, positioning, and replacement of each truck component so tested, as evidenced by tipping, losing the load, permanently deforming, fracturing, binding of the lift mechanism, or lift hydraulic system leakage, shall be cause for rejection (see 3.5.4 through 3.5.7).

4.4.10 Extended load retention test. The lift shall be operated to raise its rated load until the lifting member is in the horizontal position, and the lift tested in the following manner:

The distance between the support plate and the floor shall be measured with a height gage having an accuracy of at least ±0.001 inch. The lift shall then be allowed to stand for at least four hours. The height of the support plate shall be measured again at the end of the four hour period.

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The rated load shall then be raised to points equal to 50, 75, and 100 percent of the maximum lifting height of the lift, and the lift tested as above for at least one-half hour at each point. Failure of the lift to retain its rated load within 0.062 inch with the lifting mechanism in the horizontal position for at least four hours, failure of the lift to retain its rated load within 0.008 inch of each of the other stopping points (50, 75, and 100 percent of maximum lifting height) for at least one-half hour, or evidence of leakage shall be cause for rejection (see 3.5.2).

4.5 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or C, as specified (see 6.2.1).

5.1.1 Level A preservation.

5.1.1.1 Cleaning and drying. The lift shall be cleaned by Method C-1 and dried by procedure D-1 or D-4 in accordance with MIL-P-116.

5.1.1.2 Preservative application. After the lift has been cleaned and dried, preserve all exposed unpainted functional surfaces with an application of Type P-11 preservative. The lifting mechanism shall be actuated so that all surfaces requiring a preservative are exposed for application. Exposed, unpainted, nonfunctional surfaces shall be preserved with an application of Type P-1 or Type P-19 preservative. Caster bearings shall be preserved with Type P-11 preservative. Exposed surfaces preserved with Type P-11 preservative shall be wrapped with barrier material conforming to MIL-B-121, Grade A. Secure barrier material with tape conforming to PPP-T-60, type optional.

5.1.1.3 Unit packaging. The lift shall be packaged in accordance with Method I of MIL-P-116. To obtain economy in packing, the pump handle shall be detached and secured and the mechanism shall be secured in a closed position. Place accessory tools, lift plate adapter(s), and safety chain(s) in one or more fiberboard box(es) conforming to PPP-B-636, Class weather-resistant. Adapter hardware shall receive a double wrap of MIL-B-121 barrier material and be packaged with accessory tools. Technical data shall be packaged in accordance with Method 1C-1 of MIL-P-116 and be attached to the basic unit. Positive means shall be applied to assure that the hydraulic fluid in the hydraulic unit and reservoir shall not leak out during shipment or storage regardless of the position that the lift is placed in.

5.1.2 Level C. The lift shall be preserved in accordance with MIL-STD-1190. Positive means shall be applied to assure that the hydraulic fluid in the hydraulic unit and reservoir shall not leak out during shipment or storage regardless of the position that the lift is placed in.

5.2 Packing. Packing shall be Level A, B, or C, as specified (see 6.2.1).

5.2.1 Level A. Components comprising one complete lift shall be packed in a cleated plywood box or nailed wood box conforming to PPP-B-601, overseas type, or PPP-B-621, Class 21, respectively. The lift and packaged accessories shall be anchored, cushioned, blocked, braced and waterproofed in accordance with MIL-STD-1186 and the applicable box specification and appendix thereto. Box closure shall be in accordance with applicable specification and appendix.

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5.2.2 Level B. Level B shall be packed the same as Level A above except the wood boxes shall be domestic type in lieu of overseas type.

5.2.3 Level C. The lift shall be packed in accordance with MIL-STD-1190. Each lift shall be individually packed in a quality wooden crate and/or wood reinforced carton with pallet that provides for safe and secure stacking at least three units high during storage. All accessories shall be packed in corrugated boxes and secured inside the crate with the lift. The handle need not be packed in a corrugated box but must be secured to prevent damage during shipment.

5.3 Marking. In addition to any special markings specified in the contract or purchase order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. Bar code marking is required and shall be in accordance with MIL-STD-129 and MIL-STD-1189.

6. NOTES

6.1 Intended use. The transmission lift is intended or use by field military maintenance organizations to facilitate removal and installation of military vehicle drive line components.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of the individual documents referenced (see 2.1.1 and 2.2)
c. First article when required (3.1).
d. Painting, if different (see 3.7).
e. Selection of applicable level of preservation and packing (see 5.1 and 5.2).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, assembly instructions, and other instructions for operation and maintenance, as identified on a DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provision of the Federal Acquisition Regulation. The first article should be a preproduction sample, or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examination, test and approval of the first article.

6.5 Subject term (key word) listing.

Hydraulic
Proofload
Support plate
Casters

6.6 Sampling. Sampling for quality conformance inspection should be performed as follows. Lots should be accepted with zero defects and rejected if one or more defects are found.

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Inspection or Test	Lot or batch size	Number randomly selected for inspection
Product examination (4.3)	------	100 percent inspection
Proofload test (4.4.2)	------	test each item
Load limiting device test (4.4.3)	------	test each item
Load retention and release test (4.4.4)	1-150	8
Packaging inspection (4.5)	1-150	5

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to extensiveness of the changes.

Custodians:
Army - AL
Navy - YD
Air Force - 99

Preparing activity:
Army - AL

Review activity:
Army - AT
Air Force - 82
DLA-CS

Project No. 4910-0687

User activities:
Navy - MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-L-45047F, Lift, Vehicle Driveline Components (2000 Pound Capacity)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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